

**MASTER OF COMPUTER
APPLICATIONS (MCA)
Term-End Examination
June, 2025**

**MCS-230 : DIGITAL IMAGE PROCESSING AND
COMPUTER VISION**

Time : 3 Hours

Maximum Marks : 100

Note : *Question No. 1 is compulsory. Attempt
any **three** questions from the rest. Use of
Scientific calculator is allowed.*

1. (a) What do you mean by Images Digitization ? Explain the role of sampling and quantization in process of

digitization, with suitable example for each. 10

- (b) Elaborate on the key steps involved in a digital image processing pipeline, from the acquisition of an image to the final output. Include discussion on preprocessing, image enhancement, image transformation and the role of mathematical operations in these processes. 10

- (c) Discuss the significance of image representation in digital image processing. How do different colour models, such as RGB and CMYK, impact the way images are stored and processed ? Provide insights into the

advantages and limitations of these models. 10

(d) Compare true colour images and monochromatic images. 5

(e) What do you understand by the term “Brightness of Images” ? How is it different from the contrast of any image ? 5

2. (a) Discuss the significance of unitary transformations in the processing of 2-D signals, particularly images. Explain the properties of unitary transformations and how they preserve the inner product of vectors. Give examples of unitary transformations

applied to images, emphasizing their role in preserving information while enabling efficient processing and analysis. 7

- (b) What do you mean by Discrete Fourier Transform (DFT) ? Explain how the DFT is computed for 2-D images, and discuss the significance of understanding the frequency content of images in tasks such as image enhancement and restoration. 5

- (c) Discuss the advantages and limitations of Haar transformation compared to other transformation methods, such as

Fourier or cosine transformations. Also explain scenarios where Haar transformation is particularly well-suited and provide insights into its applications in image compression, denoising, and feature extraction. 8

3. (a) Explain the following : 8

(i) Color image enhancement techniques

(ii) Role of filtering in image enhancement

(b) What do you mean by camera configuration in influencing depth perception in computer vision systems ?

Compare and contrast the depth

perception capabilities of single-camera and multiple camera models, considering factors such as camera placement, calibration, and the impact on accurate spatial representation. 8

- (c) Briefly discuss about “Pin-Hole” camera model. 4

- 4. (a) Explain how the integration of feature extraction techniques enhances the accuracy of object detection in computer vision. 5

- (b) Compare and contrast the trade-offs associated with different edge detection algorithms. Provide insights into how

the choice of edge detection techniques can be tailored to specific applications. 7

- (c) Explain the significance of region detection in the context of image segmentation. Also discuss the challenges and importance of accurately identifying and delineating regions within an image. Provide examples of applications where region detection plays a crucial role, such as scene understanding, or object recognition. 8

5. Write short notes on any *five* of the following : 4×5=20

- (a) Supervised learning

- (b) Clustering
- (c) K-means clustering
- (d) Cosine transformation
- (e) Stereovision
- (f) Region detection

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**MASTER OF COMPUTER
APPLICATIONS (MCA-NEW)**

Term-End Examination

December, 2024

**MCS-230 : DIGITAL IMAGE PROCESSING AND
COMPUTER VISION**

Time : 3 Hours

Maximum Marks : 100

***Note : Question No. 1 is compulsory. Attempt any
three questions from the rest.***

1. (a) Differentiate between direct and indirect imaging system. 5
- (b) The dimension of an image is 5×8 inches and the frequency is 500 dots per inch in each direction. Find the number of samples required to preserve the information in the image. 5
- (c) What are the *two* types of image enhancements ? Define them with the help of suitable examples. 5

- (d) Compute the discrete cosine transform (DCT) matrix for order 2. 5
 - (e) Compare between the Continuous Wavelet Transform and Discrete Wavelet Transform. 5
 - (f) In a RGB image, the R and B components are at mid and the G component is at 1, then which colour would be seen by a person ? 5
 - (g) What are the basic properties of Linear Discrimination Function ? 5
 - (h) What are different distance measures used for clustering ? Explain with suitable examples. 5
2. (a) For a sample mobile phone with height and width resolution as 1080×1920 pixels, and the diagonal size of 5.0 inches, calculate pixel density in pixel per inch. 10
- (b) What is meant by Laplacian filter ? Using the second derivative, develop a Laplacian mask for image sharpening. 10

3. (a) What is the role of Histogram equalization in image enhancement ? Why this technique yields a flat histogram ? 10
- (b) Discuss the applications of low pass filters. 10
4. (a) Explain the types of noises based on its probability distribution. 10
- (b) Consider the coordinates of warm white (0.45, 0.4) and coordinates of deep blue (0.15, 0.2). Find the percentage of the three colours Red (R), Green (G) and Blue (B) using the information given above. 10
5. (a) Explain single link and complete link clustering with the help of a suitable example. 10
- (b) Explain Froggy's algorithm of clustering with the help of a suitable example. 10

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**MASTER OF COMPUTER APPLICATIONS
(MCA-NEW)**

Term-End Examination

December, 2022

**MCS-230 : DIGITAL IMAGE PROCESSING AND
COMPUTER VISION**

Time : 3 hours

Maximum Marks : 100

Note : Question no. 1 is **compulsory**. Attempt any **three** questions from the rest. (Use of scientific calculation is allowed)

1. (a) Differentiate between Analog Image and Digital Image. Also, explain the term Image File Format. Mention some of the frequently used image file formats. 8
- (b) How does digital image relate to signals ? Give definition of 1-D and 2-D signals and explain how a 2-D image is expressed in terms of 2-D signals. 8
- (c) Explain Discrete Fourier Transform (DFT). Your explanation must include the properties of DFT. Construct a DFT matrix of Order 2. 8

- (d) Compare Image Enhancement with Image Restoration. Explain the term “Image Degradation” and define the various types of degradations in any image. 8
- (e) What is a Discriminant Function ? Compare Linear Discriminant function with Quadratic Discriminant function. Give properties and utility of Linear Discriminant function. 8
2. (a) Compare Sampling and Quantization in context of image digitization. Also, state Shannon-Nyquist theorem and discuss its role in sampling process of any image. 7
- (b) If the physical size of a medical image is 8×8 inches and the sampling resolution is 5 cycles/mm, then how many pixels per cycle are required to have a better quality image ? Will an image of size 256×256 be enough ? 5
- (c) Explain the term “Contrast Stretching.” How does contrast stretching contribute to image enhancement ? Also, discuss the fundamental characteristics of image, i.e., contrast, brightness and intensity. 8

3. (a) Explain Haar transform, with suitable example. Compute the discrete Haar transform of a 2×2 matrix, given $N = 2$. 7
- (b) Explain how image enhancement is better in the frequency domain as compared to spatial domain. Differentiate between image smoothening filters and image sharpening filters. 8
- (c) Explain RGB model, with suitable diagram. 5
4. (a) Explain Euclidean Distance Classifier. Write the assumptions under which it turns out that optimal Bayesian classifier is equivalent to minimum Euclidean distance classifier. 7
- (b) Differentiate between the following clustering methods : 8
- (i) Hierarchical and Non-Hierarchical methods
- (ii) Agglomerative and Divisive methods
- (c) Describe the following quantities used to represent any colour : 5
- (i) Chromaticity
- (ii) Intensity

5. Write short notes on any **five** of the following : $5 \times 4 = 20$

- (a) Image representation using pixel model
 - (b) Binary images
 - (c) Intensity resolution
 - (d) Butterworth low-pass filter
 - (e) Salt and Pepper noise
 - (f) Mean filters
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No. of Printed Pages : 4

MCS-230

**MASTER OF COMPUTER
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(NEW)**

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June, 2023

**MCS-230 : DIGITAL IMAGE PROCESSING AND
COMPUTER VISION**

Time : 3 Hours

Maximum Marks : 100

Note : *Question No. 1 is compulsory. Attempt any
three questions from the rest. Use of
Scientific calculator is allowed.*

1. (a) Explain the term “Image Acquisition.”
Compare optical image processing, analog
image processing and digital image
processing. 8
- (b) Explain the term “Image transformation”.
Also, give its importance in context of
digital image processing. Write key steps
for image transformation from the spatial
domain to frequency domain. 8

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- (c) Explain Discrete Cosine Transform (DCT).
Give advantage of DCT over DFT.
Compute DCT matrix of order 2. 8
- (d) What are smoothening filters ? How do they differ from the sharpening filters ?
Briefly discuss the role of low pass filters and high pass filter in image restoration. 8
- (e) Explain Bayes' theorem with suitable example. Also, discuss the role of Bayes' theorem in Bayes' classifier. Give the properties of Bayes' classifier. 8
2. (a) What is image digitization ? Explain the role of quantization in the image digitization, with the help of an example. 7
- (b) Given, the dimension of an image as 5×8 inches² and the frequency of 500 dots per inch exists in each direction. Determine the number of samples required to preserve the information in the image. Also, discuss the term 'Pixelization error.' 5
- (c) What do you understand by the term image enhancement ? Specify the objectives of image enhancement techniques. Explain the types of image enhancement techniques with a suitable example for each. 8

3. (a) How do wavelets differ from waves ?
Explain the properties, possessed by a function to be called as wavelet. Compare wavelet transform with the Fourier transform. 8
- (b) Explain image degradation model with suitable block diagram. How noise relates to image degradation ? Explain any *one* noise model. 7
- (c) Explain HSI colour model with suitable diagram. 5
4. (a) Explain Linear Discrimination Function. How does it relate or differ from piecewise linear discriminant functions ? Write properties of Linear Discriminant Analysis (LDA). 7
- (b) Explain the following : 8
- (i) Partition based clustering
- (ii) K-means clustering
- (c) Describe the following quantities, used to represent any colour : 5
- (i) Hue
- (ii) Saturation

5. Write short notes any *five* of on the following : 4×5=20

- (a) Image representation using vector model
- (b) True colour images
- (c) Spatial resolution
- (d) Ideal low pass filter
- (e) Gaussian noise
- (f) Band Pass filter

**MASTER OF COMPUTER
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December, 2023

**MCS-230 : DIGITAL IMAGE PROCESSING AND
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Time : 3 Hours

Maximum Marks : 100

Note : *Question No. 1 is compulsory. Attempt any
three questions from the rest. Use of
scientific calculator is allowed.*

1. (a) What is an imaging system ? Compare direct imaging system with the indirect imaging system. Also, explain the term 'image digitization'. 8
- (b) Compare orthogonal transform and unitary transform. Also, discuss the properties of both the transforms. 8
- (c) Explain the term 'Wavelet' with suitable example. Give properties of wavelets. Differentiate between continuous wavelet transform and discrete wavelet transform.

8

- (d) Why do we need to transform an image from spatial domain to frequency domain ? Discuss the various categories of frequency domain filters. Also, list the filters under each category. 8
- (e) Compare unsupervised learning techniques with supervised learning techniques. Explain the different categories of supervised machine learning algorithms. Also draw block diagram for classical taxonomy of clustering methods. 8
2. (a) What is sampling ? Explain the role of sampling in image digitization, with suitable example. 7
- (b) Given a gray scale image with aspect ratio of 6 : 2 and pixel resolution of 480000 pixels. Calculate the following : 5
- (i) Dimensions of image
- (ii) Size of image
- (c) Write expression for forward and inverse orthogonal transform of an $(N \times N)$ image $f(x, y)$. Given orthogonal matrix
- $$(A) = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \quad \text{and image matrix}$$
- $$f = \begin{bmatrix} 1 & 3 \\ 5 & 7 \end{bmatrix}, \quad \text{determine orthogonal}$$
- transform and its inverse. 8

3. (a) Explain Discrete Fourier Transform (DFT). Discuss the properties of DFT. Compute the 2-D DFD of the 2×2 image

$$f(x, y) = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}. \quad 8$$

- (b) What do you understand by shifting the centre of the spectrum ? Why is it required ? Write the steps to carry out filtering in frequency domain. 7
- (c) Explain CMY and CMYK colour models. 5
4. (a) Draw a block diagram to show the learning phases of a supervised learning algorithm. Also, write steps to explain the process of applying supervised machine learning to a real world problem. 7
- (b) Explain Agglomerative Hierarchical Clustering. Write steps of general agglomerative clustering algorithm. Also discuss single-link and complete link type of agglomerative clustering. 8
- (c) Describe the following quantities, used to represent any colour : 5
- (i) Brightness
- (ii) Contrast

5. Write short notes on any *five* of the following : 4×5=20

- (a) Classification of images on the basis of attributes
- (b) Pseudo colour images
- (c) Pixel resolution
- (d) Gaussian low pass filter
- (e) Rayleigh noise
- (f) Median filters

No. of Printed Pages : 4

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June, 2024

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Time : 3 Hours

Maximum Marks : 100

Note : (i) *Question No. 1 is compulsory.*

(ii) *Attempt any **three** questions from the rest.*

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1. (a) Explain briefly the optical image processing, analog image processing and digital image processing. 5
 - (b) What is the role of sampling and quantization in the process of digitization ? 5
 - (c) Discuss the importance of image enhancement in image processing. 5

P. T. O.

- (d) Compute discrete Fourier transform of the
1 D sequence $f(x) = [1, 0, -1, 0]$. 5
- (e) What are basic properties of Fourier
transform ? Discuss in brief. 5
- (f) Explain in brief the inverse filtering
approach and its limitations in image
restoration. 5
- (g) What are the basic properties of Bayes
classifiers ? 5
- (h) Differentiate between supervised and
unsupervised learning approaches with the
help of an example. 5
2. (a) Given an image is a gray scale image with
aspect ratio of 6 : 2 and pixel resolution of
480000 pixels. Calculate the following : 10
- (i) Resolve pixel resolution to calculate
the dimensions of image.
- (ii) The size of the image.

- (b) Write short notes on the following : 10
- (i) Image processing and computer graphics
 - (ii) Image processing and machine vision
 - (iii) Image processing and video processing
3. (a) Define the concept of Histogram matching with an appropriate example. 10
- (b) Explain image degradation and also discuss the types of image degradation in detail. 10
4. (a) Write short notes on the following : 10
- (i) Gaussian Noise
 - (ii) Rayleigh Noise
 - (iii) Gamma Noise
- (b) Describe how the grey levels vary in RGB primary images that make up the front face of the colour cube. 10
5. (a) Explain K-means clustering algorithm with a suitable example. 10

(b) Write short notes on the following : 10

(i) Feature Vector

(ii) Feature Space

(iii) Decision Region and Decision Boundary

(iv) Hyperplanes and Hypersurfaces